Vivek Jude Manoj

(905)-929-8485 | vjmanoj@uwaterloo.ca | linkedin.com/in/vivekjudemanoj | github.com/viv500

Education

University of Waterloo — Waterloo, Ontario
Bachelor of Computer Science, Artificial Intelligence Specialization — *Expected Graduation: December 2027*

- Relevant Coursework: Data Structures and Algorithms, Machine Learning, Networking, Software Security

Technical Skills

Programming: Python, C/C++, Java, Go, JavaScript/TypeScript, C#
Web & APIs: React.js, Next.js, Node.js, Flask, REST & GraphQL APIs, gRPC/Protocol Buffers
Infrastructure & Cloud: Unix/Linux, AWS, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins)
AI/ML & Data: PyTorch, TensorFlow, scikit-learn, Pandas, NLP (BERT), Distributed & Event-Driven Systems

Experience

Shopify — Jan 2027 – Apr 2027
Software Engineer Intern (Incoming) — *Toronto, Ontario*

Arctic Wolf Networks — Jan 2026 – Apr 2026
Software Developer Intern - Infrastructure & Endpoint Security — *Waterloo, Ontario*

- Shipped an **AWS Dedicated Host Allocator** in **Go** and **Python**; adopted by **3+ teams** to automate EC2 provisioning, saving ∼**5 hours** per allocation.
- Provisioned **5+ macOS CI build nodes** on AWS EC2 Dedicated Hosts end-to-end, authoring custom AMIs, Jenkins pipelines, and **10+** downstream integrations for security software builds.
- Sped up **GitHub Actions** with persistent **Docker** volumes for **Bazelisk** caches on **Linux** runners, cutting **ARM64** build time ∼**40%**.

Rocket Mortgage — May 2025 – Sep 2025
Software Developer Intern - Platform Systems — *Detroit, Michigan*

- Built an **MCP-based AI DevOps automation bot** integrating GitHub, Azure DevOps, and Slack APIs to surface PR and sprint insights, cutting context-switching by **20 hrs/week**.
- Engineered **React/TypeScript** UI components for Rocket Homes' property discovery, serving **1.5M+ users**.
- Optimized cross-service data fetching with Redis caching and tuned GraphQL resolvers, reducing API latency by **35%** under production load.

AssistNow Inc. — May 2024 – Sep 2024
Software Developer Intern — *Bentonville, Arkansas*

- Built distributed **microservices** on **AWS** (ECS, Lambda, SQS) replacing a legacy monolith, improving deployment frequency by **3x** and reducing infrastructure costs by **22%**.
- Developed a parallel **C++ data ingestion pipeline** on **Linux** using **multithreading** and **lock-free queues**, processing **1M+ events/sec**.
- Built a **C++ gRPC** service bridging Node.js services via **Protocol Buffers**, cutting serialization overhead **35%**.

Projects

WatchDog - AI Scam Detection | *Python, TensorFlow, React, Flask, AWS Lambda, DynamoDB* GitHub | Demo

- Awarded **Most Promising Hack** at StarterHacks 2024; built a tool that listens to live phone calls and warns vulnerable users when scam patterns are detected, reaching **93% precision**.
- Trained a BERT-based **NLP** model with TensorFlow on AWS SageMaker to flag fraud in live speech, cutting inference latency **40%**.
- Designed distributed Flask microservices with SQS triggers to stream audio for scoring, handling **2,000+ req/min** at 95% uptime.

Explainable ECG Arrhythmia Classification | *Python, PyTorch, scikit-learn, SHAP, wfdb* GitHub | Report

- Built a reproducible **machine learning** pipeline segmenting **100,707 beats** from **44 MIT-BIH PhysioNet records** with **wfdb**, enforcing a **patient-disjoint** split with automated `pytest` checks.
- Trained **3 models** (decision tree, MLP, bidirectional **GRU** with attention) behind a shared interface, quantifying a capacity-vs-interpretability trade-off (**macro-F1 0.18 → 0.35**).
- Implemented vectorized **cost-sensitive Bayes thresholding** (5:1 miss/false-alarm costs), lifting rare-class recall **0.14 → 0.84**.